Experiment 2: Offline YOLO Detector Performance

Raw write-up for Chapter 4, Section 4.2

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Purpose of this document. This is the raw, complete write-up of Experiment 2, written before the thesis chapter; Section 4.2 of the final report is extracted from it. Every number traces to a file under `output/` named in the "Reporting rules" paragraph of its slot.

4.2.1 Simulation

Status: completed. Four dynamic recordings, 9860 left-camera frames, scored against projected simulator ground truth at a single confidence threshold.

Objective. Measure how well the project detector localises and names warehouse objects, independently of either SLAM integration, so that the dynamic filters of Experiment 3 have a known input quality.

Process. The collision mesh of every prop was projected into `cam0` at each frame to form ground-truth boxes, the project checkpoint `weight/best.pt` was run on the same frames, and predictions were matched to truth by IoU. Overlays from every recording were inspected before the aggregates were accepted (figure 2.2). Class assignment was then scored separately, on the matched pairs only, so that a naming error is never confused with a localisation error.

Datasets. The four scripted dynamic recordings of the extreme world (table 2.1). Every recording has the plain floor and 68 or 69 placed props, of which the moving ones follow scripted paths.

Table 2.1: Recordings scored in Experiment 2 (names without the `dataset_` prefix). Objects, route length and simulated span are from the dataset registry; frames are the left-camera images the detector was scored on.

Recording	Arrangement	Objects	Route [m]	Span [s]	Frames
`dynamic_nofloortexture_00_000`	dynamic_00	69	97.1	77.4	2347
`dynamic_nofloortexture_00_001`	dynamic_00	69	97.3	74.2	2236
`dynamic_nofloortexture_01_000`	dynamic_01	68	122.8	84.6	2564
`dynamic_nofloortexture_02_000`	dynamic_02	68	126.6	89.5	2713
Pooled					9860

Reporting rules. Confidence threshold 0.35, NMS IoU 0.50, matching IoU 0.50; an object counts as moving above 0.05 m s^{-1} from its pose history. Three exemptions remove objects from scoring: truth boxes under 12 px on a side are dropped, truth covered more than 70 % by a nearer object is neither hit nor miss, and a prediction more than half covered by unlabelled scenery is neither credited nor penalised. Pooled scores are micro-averages of the TP, FP and FN counts over all frames, not means of the four per-recording

scores. Class metrics are computed only on pairs associated at IoU ≥ 0.50 without requiring class agreement. Average precision is not reported because only predictions above the single threshold were retained. Source: `script/yolo_eval.py`, which writes `summary_metrics.csv`, `classification_metrics.csv` and the per-recording overlay folders under `output/yolo_eval/`; the exemption-matched recall is from `script/yolo_generalization.py`, file `generalization_metrics.csv` under `output/compare/simulation/exp-yolo-generalization/`.

Results. The detector is reliable at naming and unreliable at finding: conditional class F1 is 0.995, while it misses 41 % of the moving objects it is scored on (moving-object recall 0.591).

Table 2.2 gives the localisation scores per recording and pooled; the spread between recordings is small, moving-object recall 0.568–0.617, so the miss rate is a property of the detector and not of one scene. Figure 2.1 plots the same scores with the image area that truth and predictions cover.

Table 2.2: Offline detection against projected simulator ground truth. A detector-capability measurement, not a measurement of delivery to the online SLAM filters.

Recording	Frames	TP	FP	FN	Precision	Recall	F1	Moving recall
dynamic_00_000	2,347	9,987	4,826	5,900	0.674	0.629	0.651	0.617
dynamic_00_001	2,236	10,158	4,265	6,868	0.704	0.597	0.646	0.599
dynamic_01_000	2,564	8,142	4,226	6,064	0.658	0.573	0.613	0.568
dynamic_02_000	2,713	8,661	3,743	6,502	0.698	0.571	0.628	0.581
Pooled	9,860	36,948	17,060	25,334	0.684	0.593	0.635	0.591

Table 2.3 shows that once an object is localised its class is almost never wrong: 137 substitutions in 37,061 associated pairs, 110 of them buckets labelled as boxes (figure 2.3).

Table 2.3: Class assignment conditional on IoU association, pooled over the four recordings. Objects never localised do not enter this table; they are detection errors in table 2.2.

Class	Associated objects	Precision	Recall	F1
Bin	2,639	1.000	1.000	1.000
Box	28,076	0.996	0.999	0.998
Bucket	6,346	0.996	0.983	0.989
Macro average	37,061	0.997	0.994	0.995

Analysis. The recall figure is an upper bound, because it is computed over the easier subset that the exemptions leave. Of the 96,758 projected truth boxes, 34,476 are exempt as occluded or too small; counting every one of them as a miss moves pooled recall from 0.593 to 0.382. The training-set validation of the same checkpoint reports precision 0.958 and recall 0.882 on human-drawn boxes, so the drop to 0.684 and 0.382 on projected boxes mixes a real domain gap with a stricter scoring convention, and the two cannot be separated without relabelling a subset by hand.

Limitation, scoring convention. Projected boxes are the exact mesh silhouette, tighter than any annotator draws; a detector fitted to loose human boxes loses IoU here that represents no error on its part. This is the one difference from the training convention that no setting of the evaluator can remove.

[Ground truth never checked by hand] The projected ground truth was accepted on visual inspection of sampled overlays only. No frame has been annotated manually, so the agreement between projected and human boxes, and therefore the size of the convention effect above, is unmeasured.

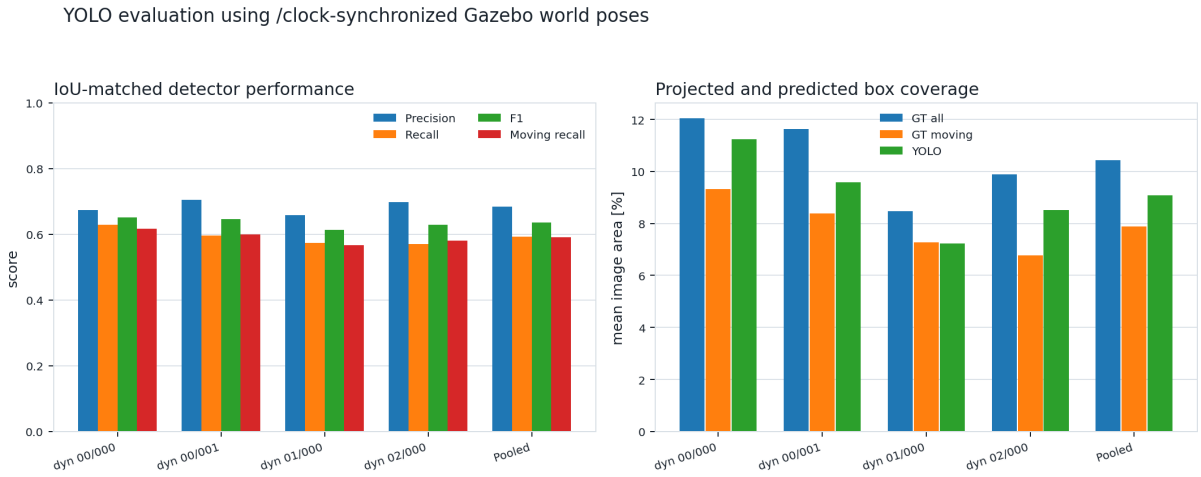


Figure 2.1: Per-recording and pooled detector scores and box-area coverage. Pooled values come from pooled counts, not from averaging the four recording scores.



Figure 2.2: Validation overlay of the kind inspected before the metrics were accepted (`dynamic_00_001`, frame 01116). Green: projected ground truth, scored. Red: predictions. Yellow: truth covered more than 70 % by a nearer object, exempt from scoring. An asterisk marks an object classified as moving from its pose history.

The miss rate bounds every downstream filter. A class mask built from these detections can at best remove the 59 % of moving objects that are found; the remaining 41 % stay in the image and reach the feature extractors of both pipelines regardless of how the mask is applied. That bound, not the mask mechanism, is the first thing Experiment 3 inherits.

The result does not bear on the online delivery anomaly of Experiment 3. The detector produces boxes on every recording offline; a filter arm that logged zero masks failed at message delivery or timestamp matching, which is a different mechanism from detector capability.

Summary. The detector names what it finds almost perfectly and finds about six moving objects in ten, or fewer than four in ten when exempt objects are counted. Precision 0.684 means one prediction in three is a false box. These are the input limits of the dynamic filters tested in Experiment 3.

Future plan. Annotate a stratified subset of frames by hand to measure the projected-versus-human box convention and separate it from the domain gap. Re-score on the realistic warehouse recordings, where the moving objects are pallet jacks on logistics jobs rather than scripted props. Add object velocity from consecutive detections so that a mask can target moving objects rather than object classes.

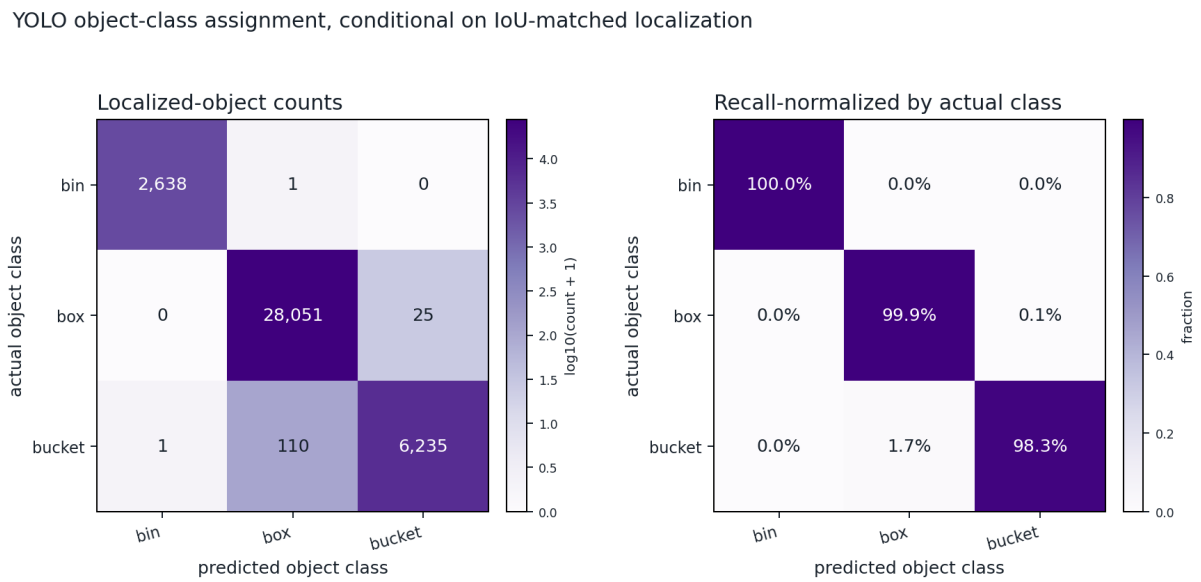


Figure 2.3: Class confusion conditional on class-agnostic IoU association. Rows are true classes, columns predicted. Left: counts on a logarithmic colour scale. Right: normalised within each true class. Background is excluded by construction.

4.2.2 Real robot

Status: planned. Real imagery exists; no real ground-truth boxes do.

Reason. Projected ground truth needs object poses and meshes, which the real site does not provide. The three real recordings hold 2562–3405 usable stereo pairs each, so the detector can be run on them, but nothing can score it until frames are annotated by hand.

Next action. Annotate a stratified frame subset of `dataset_real_000` to `dataset_real_002` across the three illumination levels of the test site, run the detector at the thresholds of section 4.2.1, and report the same metrics. The same annotated subset calibrates the projected and manual box conventions against each other for the simulation slot.